Multilabel Aerial Image Classification With Unsupervised Domain Adaptation

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***Abstract*— Deep learning (DL) methods are promising for the multilabel aerial image classification (MAIC) task. However, current DL methods face a common problem: the need for large multilabeled datasets. Collecting and annotating raw aerial image datasets can be extremely time- and labor-consuming. To address this concern in MAIC, domain adaptation (DA) provides a novel solution by transferring the knowledge learned from a label-rich dataset (i.e., the source domain) to a label-scarce dataset (i.e., the target domain), while current DA models are mainly designed for single-labeled tasks. In this article, we propose a novel end-to-end MAIC model based on DA techniques, named DA-MAIC. To the best of our knowledge, this article for the first time integrates DA to tackle the label scarcity problem in the MAIC task. Specifically, the proposed DA-MAIC is composed of two main parts: the image classifier and the domain classifier. The image classifier captures task-discriminative features based on the graph convolutional network (GCN) to predict multiple image labels; and the domain classifier extracts domain-invariant representations, which mitigates the domain shift between two underlying distributions. We extensively evaluate the proposed DA-MAIC from different perspectives on three benchmark datasets, including the commonly used UCM dataset, the high-resolution AID dataset, and the recently proposed DFC15 dataset. Both quantitative and qualitative results support that the proposed DA-MAIC can generalize the source domain knowledge to new scenarios and substantially improve the classification performance on the target domain task.**

***Index Terms*— Aerial images, domain adaptation (DA), multilabel image classification.**

NOMENCLATURE

Summary of Major Mathematical Notations

Notations	Mathematical Meanings
$D_s = (X^s, L)$	Source domain with n_s annotated aerial images $X^s = \{x_1^s, x_2^s, \ldots, x_{n_s}^s\}$.
$D_t = (X^t)$	Target domain with n_t aerial images $X^t = \{x_1^t, x_2^t, \ldots, x_{n_t}^t\}$.
$\mathbf{f}_i$	m-dimensional feature vector of the image x_i, $\mathbf{f}_i \in R^m$.
L	Finite set of c labels (tags or objects) $\{l_1, l_2, \ldots, l_c\}$.
$\mathbf{l}_i$	d-dimensional feature vector of label l_i, $\mathbf{l}_i \in R^d$.
$\mathbf{G}$	Label-level feature vectors from label encoder module, $\mathbf{G} \in R^{c \times m}$.
$\mathbf{y}_i$	Annotations for image x_i, $\mathbf{y}_i = \{y_i^1, y_i^2, \ldots, y_i^c\}$.
$\hat{\mathbf{y}}_i$	Predicted result of the image x_i, $\hat{\mathbf{y}}_i \in R^c$.

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I. INTRODUCTION

WITH the advances of remote sensing technologies, it is getting easier to obtain large volumes of aerial images with high resolutions and abundant contents. Such images are widely used in real projects, such as geomorphological monitoring [1], urban planning [2], and ecological environmental management [3]. To this end, providing comprehensive studies for these aerial images is of great importance. Current studies mainly concentrate on single-label aerial image classification (SAIC), which can predict only one label for a given image [4]. However, many aerial images inherently contain multiple objects, and SAIC is not preferred for these image understandings in real-world applications. In contrast, the emerging multilabel aerial image classification (MAIC) task has attracted increasing attention.

Different from the traditional SAIC task, MAIC aims to recognize more fine-grained objects for a given image. As shown in Fig. 1(a), several objects are labeled in the illustrative aerial image, including grass, trees, sand, and water. In this way, MAIC can provide a more detailed understanding of the aerial images and, therefore, is more suitable for real-life scenarios. Recent advances of MAIC models mainly rely on extracting task-specific aerial image features (such as DELTA [5], TResNet [6], and KSSNet [7]) and predicting multiple objects (such as recurrent neural networks (RNNs) [8], [9] and MLGCN [10]).

Although a wide variety of MAIC approaches have significantly improved the state of the arts, most existing studies succeed based on the assumption that abundant annotated images are available during the model training. While a practical challenge still exists for such models, there is often not enough publicly available aerial data with annotations

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(a) (b)

Fig. 1. Multilabel aerial image examples from two datasets. (a) Image (with grass, sand, trees, and water objects) from the UCM dataset. (b) Image (with grass, trees, and water objects) from the AID dataset.

to train the huge number of parameters needed in current MAIC models. As a result, aerial image datasets with fewer annotations make most current MAIC models less attractive in practice.

To tackle this concern, our motivation is to build the bridge between existing annotated datasets and new datasets with less information available for the MAIC task. This idea is inspired by the domain adaptation (DA) technique, which transfers the knowledge learned from a label-rich dataset (i.e., the source domain) to a label-scarce dataset (i.e., the target domain). For instance, in Fig. 1, the images from two datasets are different but similar in objects. Many DA-based models were proposed recently, such as adaptive feature norm (AFN) [11], discriminative adversarial domain adaptation (DANN) [12], and the maximum classifier discrepancy (MCD) model [13]. However, such existing DA-based models are designed for single-labeled tasks and have not been investigated for multilabel image learning scenarios yet. Directly utilizing DA models on the MAIC task can suffer from suboptimal performance as they cannot handle the complicated correlations between multiple labels. Label correlations imply the co-occurrence of different labels in the same image. For example, the presence of label water indicates a great probability of label trees. As the objects normally coexist in the real world, a key challenge for the MAIC task is to model the label correlations. In this article, to introduce the DA method to the MAIC task, we introduce label dependencies proposed in the graph convolutional network (GCN) [14]. GCN can learn graph typologies directly via the information propagation between nodes. With the label knowledge from GCN, the improved DA can provide a new solution for the label scarcity problem in the MAIC task.

In this article, we study the novel research problem of employing DA in the MAIC task. A new method for the MAIC task is proposed, named DA-MAIC. DA-MAIC is composed of two main components: the image classifier and the domain classifier. To capture multiobject task-discriminative features, the multilabel classifier utilizes the GCN to build the label relationship in the label space. To mitigate the domain shift between two distributions, the domain classifier aims to discriminate whether each image belongs to the source or the target domain.

The main contributions of this article are given as follows.

1) We introduce DA to the MAIC task, representing a novel research problem. In this problem, a model is trained on one labeled image dataset (the source domain) and tested on the other unlabeled image dataset (the target domain) with different feature distributions. To the best of our knowledge, it is the first time that DA is integrated to tackle the label scarcity problem in the MAIC task.
2) To allow the DA techniques for handling the MAIC task, the proposed DA-MAIC is well designed. A GCN-based multilabel image classifier extracts the complementary information of label correlations. A domain classifier is utilized to reduce the distribution mismatch between the source and target domains.
3) The experiments are designed to evaluate both the effectiveness and the feasibility of the proposed DA-MAIC on three datasets. Quantitative and qualitative results support that the proposed method can help generalize the knowledge from the source domain to the target domain.

II. Related Work

Recently, MAIC has become an active research topic, as it provides the annotations for follow-up aerial computer vision tasks, such as object detection [15], action recognition [16], and image segmentation [17], [18].

A. Overview on MAIC

In the past few years, under increasing space missions, such as city management, urban mapping, and disaster monitoring, large volumes of satellite aerial images containing detailed information in high resolutions have been accumulated and accessible for the academic research [19]. Extensive attention has been stressed to annotating multiple objects of those images.

Early MAIC methods behaved in a traditional way that tends to require a high computational cost. Representative models include stacked autoencoder methods [20], SVM-based models [21], and conditional random fields model [22]. Modern MAIC approaches achieved significant performance with the deep learning (DL) frameworks. Various CNN-based structures were designed for real-world applications, including unmanned aerial vehicles (UAVs) images [23], Sentinel-2 imagery acquired [24], and so on. In addition, several specific mechanisms were introduced in MAIC models to help improve the classification performance. The bidirectional RNN architecture predicted multiple labels in a sequential way [8], [25]. The Mahalanobis distance-based K-medoids approach [26] segmented the image into the instance bags. The global relationships among aerial images [27] and the local relationships among spatial regions [28] were also researched and exploited as important information during the label predicting phase.

However, those models have a strong dependence on the accessible labeled dataset during training and learning phases. There are cases where training data are expensive or difficult to collect and annotate. Especially in the remote sensing ecosystem, aerial images are usually collected in certain ways under specific environments. The annotation process for

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those images can be time- and labor-consuming. Lacking an annotated dataset leads to the model being hard to generalize well.

B. Overview on DA

In recent years, DA provides an alternative way for image classification by taking advantage of the knowledge from a label-rich domain (i.e., source domain) to a label-scarce domain (i.e., target domain). Large volumes of studies have been conducted, focusing mainly on vision-based classification tasks [29]–[31].

Among these studies, one crucial problem to overcome is the unavoidable domain shifts across distributions of the source domain and the target domain. To this end, the key point of most existing methods is to extract image representations that are not only task-specific but also domain-invariant. Typical models aligned the feature distributions by minimizing some statistical discrepancy between two domains, such as H-divergence, H_H-divergence, and maximum mean discrepancy (MMD) [32]. More recently, adversarial DA, which generates representations with an adversarial objective between the domain discriminator and the feature generator, has held strong momentum in this field [33], [34]. MM-SADA combined adversarial training with multimodal self-supervision, using both RGB and optical flow modalities [35]. GCAN jointly modeled three types of information, namely, data structure, domain label, and class label in a unified framework to bridge source and target domains [36]. After revealing that those task-specific features with larger norms are more transferable, AFN achieved domain alignment by a progressive feature-norm enlargement for each sample across domains [11].

All the above methods are designed based on single-label vision tasks, where an image matches to only one label (object). However, in real-world applications, images can contain abundant information and often be associated with multiple labels. We propose a novel solution for multilabel image classification by incorporating DA.

III. Methodology

Our objective is to investigate the effectiveness of exploring DA in the multilabel image classification task. In this section, we will discuss the detailed components of the proposed framework.

A. MAIC Task Definition

In the MAIC task with DA, there are two domains for the model training and testing. In the training phase, the annotated source domain and the nonannotated target domain are fed into the whole model. In the testing phase, the trained model is directly conducted on the testing set of the target domain. The source domain $D_s = (X^s, L)$ contains samples from the annotated dataset, and the target domain $D_t = (X^t)$ contains samples without annotation. DA models are designed in a scenario where annotated target examples are not accessible. In this article, we mainly consider the Vanilla Setting [11] of DA. Under this setting, the source and target domains share the common label space, denoted as a finite set of c labels (tags or objects) L, $L = \{l_1, l_2, \ldots, l_c\}$. DA is built when the distributions of the source and target domains in the identical label space are different but similar, making transfer feasible [37].

For the given annotated aerial image dataset in source domain D_s, the aerial image set of n_s images can be represented by $X^s = \{x_1^s, x_2^s, \ldots, x_{n_s}^s\}$. For an image $x_i^s \in X^s$, the annotations are represented as a binary vector $\mathbf{y}_i = \{y_i^1, y_i^2, \ldots, y_i^c\}$, where $y_i^k = 1$ if x_i^s is labeled with the kth label l_k, $k \in [1, 2, \ldots, c]$; otherwise, $y_i^k = 0$. For the aerial image dataset in target domain D_t, the aerial image set of n_t images can be represented by $X^t = \{x_1^t, x_2^t, \ldots, x_{n_t}^t\}$.

The goal of our MAIC task with DA is to build an adaptable multilabel image classifier that can map the given target image x_i^t to the output label vector $\mathbf{y}_i$, $f : x_i^t \rightarrow \mathbf{y}_i$. An unsupervised DA learning algorithm is then provided with a labeled source domain D_s and an unlabeled target domain D_t. In this way, DA-MAIC constructs an unsupervised solution where the annotations of the target domain are not essential during the model training. For easy reference, Nomenclature lists major notations in this article along with their mathematical meanings.

B. DA-MAIC Framework

The proposed framework is illustrated in Figs. 2 (for the training phase) and 4 (for the testing phase). As illustrated in Fig. 2, during the model training phase, there are mainly two components in the proposed model: the image classifier and the domain classifier. The image classifier extracts task-specific features for multilabel prediction with the image encoder module and the label encoder module. The structure of the label encoder is shown in Fig. 3. The domain classifier learns the common representations to reduce the domain gap. As illustrated in Fig. 4, during the model testing phase, the trained model performs the multiple label prediction on the target testing set.

To allow the DA techniques for handling the multilabel image learning task, the proposed DA-MAIC is well designed with two specific steps. First, for the image feature learning in Fig. 2, the step of Image prediction in the image classifier can generate multiple candidates for the given image. Second, the Label encoder in Fig. 3 is designed to extract the complementary information of label correlations for the multiple labels predicting.

C. Image Classifier

Based on the above discussion, in this article, to introduce DA into the MAIC task, the image classifier integrates both image features and label correlation features. Specifically, image features are directly extracted by the pretrained backbone in the image encoder module, and the label dependency features are learned by the label encoder.

First, the image encoder module aims to learn image features by the CNN-based model. For the given raw aerial images, the pretrained ResNet network is used as a backbone

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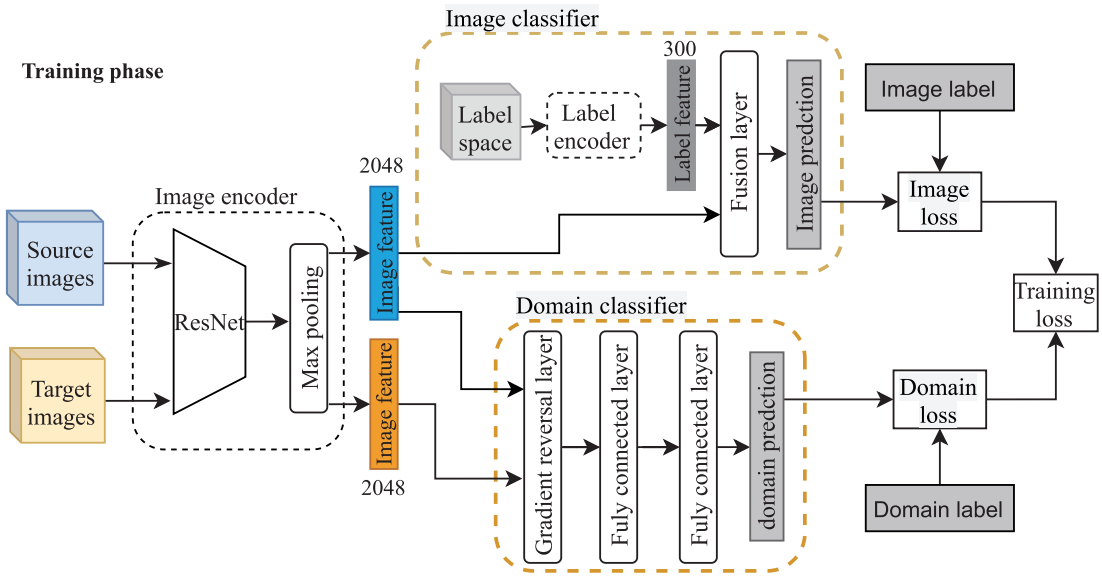


Fig. 2. Flowchart of the proposed DA-MAIC framework at the training phase.

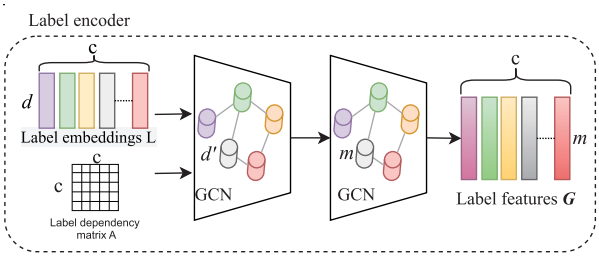


Fig. 3. Detailed structure of the label encoder.

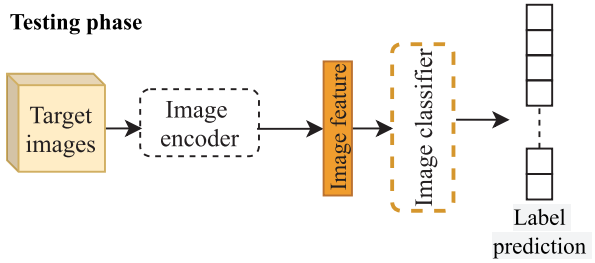


Fig. 4. Flowchart of the proposed DA-MAIC framework at the testing phase.

to learn the preliminary image features. Taking raw image x_i as the input, the output of the backbone is with a dimension of 2048 × 14 × 14, followed by a max-pooling layer to get the image-level feature $\mathbf{f}_i \in R^m$, $m = 2048$.

Second, the label encoder module extracts label correlation features with the GCN module. In the MAIC task, correlations among labels can provide supplemental information during label predicting. For example, if label sea exists in the image, there is a high probability that label sand coexists in the same image. In this way, label dependencies are considered as the crucial knowledge during the predicting phase.

In this article, we first represent all label nodes in the label space L as d-dimensional feature vectors $\mathbf{L}$, where each label node $l_i \in L$ is denoted as $\mathbf{l}_i \in R^d$. Inspired by MLGCN [10], for the common label space L, we build the label dependency matrix A in the form of conditional probability. Specifically, for label l_i and l_j, $P(l_j|l_i) \in \mathbf{A}$ denotes the probability of existence of label l_j when label l_i appears. GCN can directly deal with non-Euclidean structure with label representations $\mathbf{L} \in R^{c\times d}$ and the label dependency matrix $\mathbf{A} \in R^{c\times c}$ as inputs. In the layer a, the input node feature $\mathbf{L}^a$ is updated into the output node feature $\mathbf{L}^{a+1}$ by

$$\mathbf{L}^{a+1} = h\left(\mathbf{A}\mathbf{L}^a\mathbf{W}^a\right) \tag{1}$$

where $\mathbf{W}^a \in R^{d\times d'}$ is a weight transformation matrix to be trained and $h(\cdot)$ is the nonlinear activate function (i.e., LeakyReLU in this article). The label correlation feature can be updated by integrating information from other nodes in GCN layers. We denote the output of GCN-based label encoder as $\mathbf{G} \in R^{c\times m}$.

Third, the fusion layer predicts multiple labels by considering features of images and the dependencies of labels. Specifically, for the input image x_i, the predicting probabilities $\hat{\mathbf{y}}_i \in R^c$ on each label are calculated by the dot-product similarity between the image feature $\mathbf{f}_i \in R^m$ and the label correlations $\mathbf{G} \in R^{c\times m}$, i.e.,

$$\hat{\mathbf{y}}_i = \text{sigmoid}\left(\mathbf{G}\mathbf{f}_i^{\mathrm{T}}\right). \tag{2}$$

For image x_i, each label l_j has both the corresponding final prediction $\hat{y}_i^j$, indicating the confidence of label l_j in image x_i, and the ground-truth $y_i^j \in \mathbf{y}_i$, which represents the existence of the label l_j. If the score $\hat{y}_i^j$ is above 0.5, we set the label l_j as positive for the image x_i. We utilize the multilabel cross-entropy loss function to optimize the model as

$$\mathcal{L}_C = \frac{1}{c}\sum_{j=1}^{c}\left[y^j \log\left(\hat{y}^j\right) + \left(1 - y^j\right)\log\left(1 - \hat{y}^j\right)\right] \tag{3}$$

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where c is the size of the label set L. y^j and $\hat{y}^j$ are the jth element of $\mathbf{y}$ and $\hat{\mathbf{y}}$, respectively, $j \in [1, 2, \ldots, c]$.

D. Domain Classifier

1) Domain Divergence: One challenge in DA tasks is the divergence between source and target domains. Most existing methods built the notion of distance from the source domain to the target domain, Maximum mean feature norm discrepancy [11], MMD [32], and $\mathcal{H}$-divergence [12], to name a few. These methods are justified as the source risk can provide good indications for the target risk when their features maps are similar. In this article, we focus on the empirical $\mathcal{H}$-divergence [38] between source and target domains

$$\hat{d}_{\mathcal{H}}(D_s, D_t) = 2\left(1 - \min_{\alpha\in\mathcal{H}}\left[\frac{1}{n_s}\sum_{j=1}^{n_s} I\left[\alpha\left(\mathbf{f}_j\right) = 0\right] + \frac{1}{n_t}\sum_{j=1}^{n_t} I\left[\alpha\left(\mathbf{f}_j\right) = 1\right]\right]\right) \quad (4)$$

where $\mathbf{f}$ means the image representations extracted by the pretrained backbone. D_s is the source domain distribution (with n_s samples), and D_t is the target domain distribution (with n_t samples). The hypothesis class $\mathcal{H}$ is a (discrete or continuous) set of binary classifiers $\alpha(x) : x \to 0, 1$. $I[\cdot]$ is the indicator function that is 1 if true and 0 otherwise. For more detailed definition of the empirical $\mathcal{H}$-divergence, we refer interested readers to [29].

2) Domain Classifier: The domain classifier is consists of one gradient reversal layer (discussed in detail in Section III-E) and two CNN layers, as shown in Fig. 2.

In our MAIC task, we utilize the binary classifier H to learn the domain features of images as

$$H(\mathbf{f}; w, a) = \text{sigmoid}\left(w^{\mathrm{T}}\mathbf{f} + a\right) \quad (5)$$

where $\mathbf{f}$ represents the image feature, and the parameters of vector $w \in R^m$ and scalar a are to be learned.

We utilize the cross-entropy loss for optimization as

$$L_H(H(\mathbf{f}_i), d_i) = d_i \log\frac{1}{H(\mathbf{f}_i)} + (1 - d_i)\log\frac{1}{(1 - H(\mathbf{f}_i))} \quad (6)$$

where $H(\mathbf{f}_i)$ is the output of the binary classifier H for image i, and d_i is the ground truth of the domain labels, which tells the domain of sample $\mathbf{f}_i$ as

$$d_i = \begin{cases} 0, & \text{if } \mathbf{f}_i \in D_s \\ 1, & \text{if } \mathbf{f}_i \in D_t. \end{cases} \quad (7)$$

The domain label for one image is a binary value. We set the domain label as 0 for all source images and 1 for all target images. In this way, the domain label and the image label are different: the domain label tells which domain the image comes from; the image label tells objects of this image. Therefore, the domain divergence can be optimized by the classifier H as

$$\mathcal{L}_H = \hat{d}_{\mathcal{H}}(D_s, D_t) = 2\left(1 - \min_{\alpha\in\mathcal{H}}\left[\frac{1}{n_s}\sum_{j=1}^{n_s} L_H(H(\mathbf{f}_i), d_i) + \frac{1}{n_t}\sum_{j=1}^{n_t} L_H(H(\mathbf{f}_i), d_i)\right]\right). \quad (8)$$

E. Training Objective

To learn a model that can generalize well from the annotated domain to the nonannotated domain, we incorporate a DA component into the multilabel image classification. In consequence, the complete optimization objective of the MAIC task is

$$\mathcal{L} = \mathcal{L}_C + \lambda\mathcal{L}_H \quad (9)$$

where $\lambda \in R$ is the hyperparameter in the model training. To learn the domain-invariant features $[\min \hat{d}_{\mathcal{H}}(D_s, D_t)]$, we need to maximize the loss of domain regression $L_H(H(\mathbf{f}_i), d_i)$ during model training as

$$\min \hat{d}_{\mathcal{H}}(D_s, D_t) \Leftrightarrow \max\min_{\alpha\in\mathcal{H}} \times\left[\frac{1}{n_s}\sum_{j=1}^{n_s} L_H(H(\mathbf{f}_i), d_i) + \frac{1}{n_t}\sum_{j=1}^{n_t} L_H(H(\mathbf{f}_i), d_i)\right]. \quad (10)$$

The whole model acts in an end-to-end way during the training phase, and the image classifier and the domain classifier $H(\mathbf{f}; w, a)$ are competing against each other in an adversarial way. Specifically, the image classifier aims to classify source instances precisely but hinders the domain classifier distinguish whether the instance is from the source or target space. To achieve the model training, as in [29], a gradient reversal layer is introduced into the domain classifier. The gradient reversal layer can optimize the mapping to maximize the domain classifier loss directly by multiplying the backpropagation gradient with a certain negative constant (-1).

IV. Experiments and Results

To demonstrate the feasibility and effectiveness of the proposed DA-MAIC, experiments are designed both quantitatively and qualitatively. On the one hand, quantitative results show the scores of the proposed model together with other competitors on several metrics. On the other hand, qualitative results demonstrate the visualization analyses of the model under different ablation study settings. The following subsections discuss the details of experiments in Section IV-A and analyses of results in Sections IV-B–IV-D.

A. Experiment Details

1) Evaluation Metrics: Precision, recall, and F-score are three commonly used metrics in classification tasks. Following the previous work in the MAIC task, besides mean Average

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TABLE I
NUMBERS OF IMAGES AND CLASSES IN MULTILABEL AERIAL IMAGE DATASETS

Dataset	Image	Class	Training	Testing
UCM	2, 100	17	1, 680	420
AID	3, 000	17	2, 400	600
DFC15	3, 342	8	2, 674	668

Precision (mAP), the performance can be valued on both sample- and label-based angles [39]. Here, a sample consists of an aerial image and its corresponding multiple labels. The label-based metrics demonstrate the performance of networks from the perspective of each object label. In this way, we calculate the average example-based precision (EP), recall (ER), F-scores (EF_1 and EF_2), and the average label-based precision (LP), recall (LR), and F-scores (LF_1 and LF_2) as metrics. Also, following the previous work, we compute the mean F-score $MF_1 = (EF_1 + LF_1)/2$.

2) Implementation Details: The details of the model components are given as follows. The image encoder module utilizes the pretrained Resnet-101 (with the output size of $2048 \times 14 \times 14$ from the "conv_5x" layer) as the backbone. Then, a max-pooling layer (with the kernel size of 7×7) is stacked for feature extracting. There are two stacked GCN layers in the label encoder module, with dimensions of 1024 and 2048, respectively. The fusion layer is a dot product operation layer with the output of the label predictions. On the other hand, the domain classifier consists of one gradient reversal layer and two CNN layers (with the output sizes of 100 and 2, respectively). The input of the domain classifier is the image features (extracted by the image encoder), and the output is the binary domain prediction (calculated by the sigmoid operation).

We select the hyperparameters of our model via grid search according to the metrics on the testing set. Specifically, we select the learning rate among $\{0.0005, 0.001, 0.003, 0.005\}$, the batch size among $\{8, 16, 32\}$, and the λ among $\{0.5, 1, 2\}$. We utilize the SGD as the optimizer of the network and LeakyReLU as the nonlinear activation function. The network is trained for 20 epochs in total. All experiments are performed on NVIDIA GeForce RTX GPU and implemented in Python using the PyTorch framework.

3) Datasets: We employ three multilabel aerial image datasets, UCM [40], AID [9], and DFC15 [25]. UCM and AID share the same label space with 17 labels. AID and DFC15 are both with high resolutions. The numbers of images and classes in each dataset are shown in Table I. Besides, the statistics of each class in total, training, and testing sets are shown in Fig. 5. Details about these three datasets are given as follows.

Rebuilt from the single-labeled UC Merced Land Use Dataset [41], the UCM multilabel dataset is annotated with multiple tags based on the visual inspection. As a commonly used dataset in MAIC tasks, there are totally 2100 samples and 17 kinds of labels in UCM. Each sample has a spatial resolution of one foot with 256×256 pixels. The label space

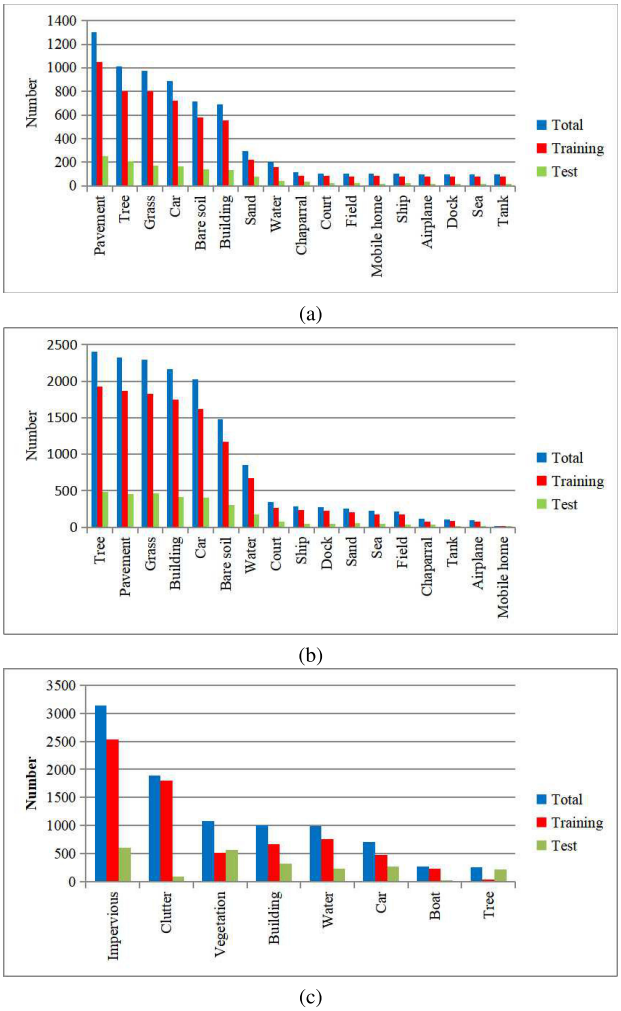


Fig. 5. Image statistics corresponding to each label in the total dataset (blue), training dataset (red), and testing dataset (green). (a) UCM multilabel dataset. (b) AID multilabel dataset. (c) DFC15 multilabel dataset.

consists of airplane, sand, pavement, buildings, cars, chaparral, court, trees, dock, tank, water, grass, mobile-home, ship, bare-soil, sea, and field. In this work, we randomly sample 80% of images evenly from every category for the model training.

The AID multilabel dataset [9] is extended from the public AID dataset [42], including a scale of 3000 aerial images with manually annotated multiple labels. Images in the AID multilabel dataset are with the size 600×600 and high spatial resolutions varying from 0.5 m/pixel up to 8 m/pixel. The 17 labels of this dataset are the same as that of the UCM multilabel dataset.

The DFC15 multilabel dataset [25] was published and first used in the 2015 IEEE GRSS Data Fusion Contests. It is more challenging with an extremely higher spectral resolution of 5 cm with respect to UCM. The label set includes eight objects, impervious, water, clutter, vegetation, building, tree, boat, and car. In our task, as DA is based on the Vanilla Setting [11], we adopt the common six classes in UCM and AID datasets in our experiments, including water, grass, building, tree, ship, and car. There are totally 3342 samples in DFC15. 80% of

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TABLE II

DA RESULTS FROM UCM TO AID (%). WE COMPARE WITH SEVERAL EXISTING METHODS FOR TWO TYPES OF TASKS, INCLUDING THE MAIC TASK AND THE DA TASK. FOR OUR RESULTS, WE FIRST REPORT THE MODEL PERFORMANCE (DA-MAIC), AND THEN, WE SHOW THE RESULTS OF THE PROPOSED MODEL WITHOUT THE DOMAIN CLASSIFIER COMPONENT (-DC) AND WITHOUT THE LABEL DECODER COMPONENT (-GCN), RESPECTIVELY

Type	Methods	EP	ER	EF_1	EF_2	LP	LR	LF_1	LF_2	MF_1	mAP
MAIC	DELTA [5]	51.02	34.65	41.27	37.03	32.46	15.13	20.64	16.94	30.96	31.826
	KSSNet [7]	58.84	24.87	34.97	28.12	7.12	9.22	8.03	8.70	21.50	26.448
	TResNet [6]	54.80	36.09	43.52	38.73	15.97	17.66	16.77	17.29	30.15	33.982
	ASL [43]	78.79	44.99	57.28	49.21	53.27	31.06	39.24	33.89	48.26	39.787
DA	CDAN [44]	80.94	44.24	57.21	48.65	48.92	21.72	30.08	24.44	39.43	39.305
	MCD [13]	80.69	35.31	49.12	39.79	30.67	18.11	22.77	19.73	31.12	38.003
Ours	**DA-MAIC**	**82.94**	**65.08**	**72.93**	**68.01**	**59.08**	**36.66**	**45.25**	**39.68**	**59.09**	**51.206**
	-DC	71.48	35.25	47.22	39.23	39.86	24.96	30.70	26.98	37.21	46.325
	-GCN	75.18	53.34	62.40	56.63	25.85	23.23	24.47	23.71	35.16	49.329

them are selected as the training set randomly, and the rest is as the testing set.

B. Quantitative Analysis

We consider the underlying questions during the whole process of integrating DA into the MAIC task. These questions progressively form the whole problem-solving process and can help verify the effectiveness and feasibility of the proposed model. The questions are given as follows.

1) How does the DA improve the classification performance of MAIC?
2) Does the spatial resolution of the source domain have an effect on the model effectiveness?
3) Can DA work with various existing MAIC methods?

To check answers to these questions, we evaluate different ways for experiments and their corresponding results. For the first question 1), to verify the performance of our proposed model, we compare with the state-of-the-art models in MAIC tasks and two existing DA models. Also, to prove the effectiveness of the DA component, we carry out the ablation study for our model. Second, to investigate question 2), we conduct the transfer between benchmark datasets with similar and disparate spatial resolutions, respectively. UCM is with the lower image resolution, and AID and DFC15 are with high spatial resolution. Specifically, we first evaluate DA-MAIC on bidirectional DA between UCM and AID datasets. The results of the two directions are illustrated in the bar charts for comparison. In addition, experiments on DFC15 (as the target domain) are designed with AID and UCM as the source domain, respectively. Third, to answer question 3), we combine the DA component with several representative MAIC methods and report the experimental results. All the results of the designed experiments are illustrated and discussed in the following.

1) DA From UCM to AID: To evaluate our model comprehensively, a quantitative analysis is conducted for both existing MAIC methods and current DA methods. As shown in Table II, besides our results, there are two types of state-of-the-art models: MAIC and DA. The experiments are conducted under the circumstance where the UCM acts as the training set and AID is the testing set. To show the effectiveness of the proposed DA-MAIC for the MAIC task, we compare with four existing methods for the multilabel image classification task (MAIC type), including DELTA [5], KSSNET [7], TResNet [6], and ASL [43]. As can be seen from Table II, the proposed DA-MAIC model surpasses all competitors on the target AID multilabel dataset. Specifically, DA-MAIC increases MF_1 and mAP scores by 10.83% and 11.419%, respectively, in comparison with ASL (state-of-the-art). In addition, our model outperforms state-of-the-art not only in sample-based indexes but also in label-based scores. Compared to KSSNet, which resorts to employing the GCN framework for exploring label correlations, our model achieves an improvement of 24.758% on mAP. Comparisons with current MAIC models demonstrate that our model can be more adaptable for the target dataset.

Also, we report results on comparison with two current DA methods, including CDAN [44] and MCD [13]. It can be observed intuitively that the proposed DA-MAIC model outperforms existing DA-based models significantly. For example, improvements of 11.901% and 13.203% for the mAP score are obtained in comparison with CDAN and MCD, respectively.

For the proposed DA-MAIC model, an ablation study is carried out to evaluate the effectiveness of each component of the proposed model: the GCN component and the domain classifier component. We first list the model performance (DA-MAIC itself), and then, we show the results of the proposed model without the DA component (-DC) and the GCN component (-GCN) separately. It is clear that, when removing the domain classifier, the performance drops sharply. Specifically, the EP decreases 11.46% from 82.94% to 71.48%, and the ER has a significant decrease of 29.83% from 65.94% to 35.25%. Without the domain classifier component, the mAP of -DC drops from 51.206% to 46.323%, indicating the effectiveness of the DA part. In addition, without the GCN component, the mAP of -GCN decreases slightly by 1.877%, indicating the effectiveness of the GCN component as providing complementary information in the label classifier. The ablation study supports the necessity of each component for the DA-MAIC model in the new MAIC task.

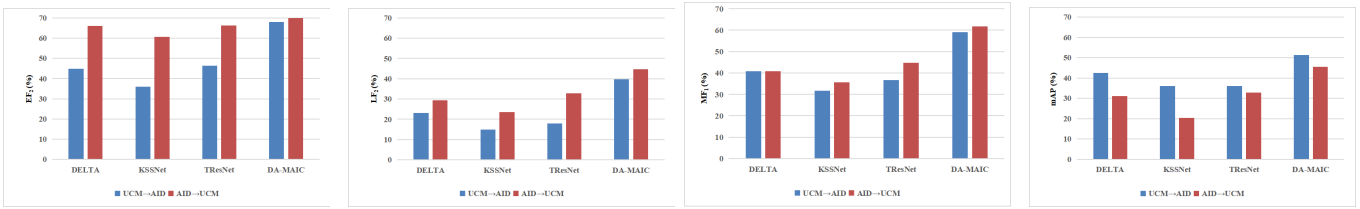
2) DA From AID to UCM: To explore the answer to question 2), besides transferring from the low-resolution UCM

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TABLE III

DA RESULTS FROM AID TO UCM (%). WE COMPARE WITH SEVERAL EXISTING METHODS FOR TWO TYPES OF TASKS, INCLUDING THE MAIC TASK AND THE DA TASK. FOR OUR RESULTS, WE FIRST REPORT THE MODEL PERFORMANCE (DA-MAIC), AND THEN, WE SHOW THE RESULTS OF THE PROPOSED MODEL WITHOUT THE DOMAIN CLASSIFIER COMPONENT (-DC) AND WITHOUT THE LABEL DECODER COMPONENT (-GCN), RESPECTIVELY

Type	Methods	EP	ER	EF_1	EF_2	LP	LR	LF_1	LF_2	MF_1	mAP
MAIC	DELTA [5]	47.48	54.69	50.83	53.08	15.23	28.79	19.92	24.44	35.38	26.160
	KSSNet [7]	50.79	55.31	50.79	55.31	12.32	24.89	16.48	20.67	33.64	20.890
	TResNet [6]	48.45	69.59	57.13	64.00	29.60	34.04	31.66	33.05	44.40	37.034
	ASL [43]	38.26	79.25	51.61	65.27	16.15	30.46	21.11	25.87	36.36	12.330
DA	CDAN [44]	71.80	48.56	58.26	52.09	46.33	26.10	34.00	28.89	46.13	41.345
	MCD [13]	73.37	31.70	45.59	36.33	47.39	16.61	26.36	19.81	35.98	40.571
Ours	**DA-MAIC**	**74.08**	**80.65**	**77.23**	**79.25**	**48.76**	**43.78**	**46.13**	**44.69**	**61.68**	**45.506**
	-DC	66.69	51.97	58.55	54.44	36.23	25.01	29.85	26.80	44.20	39.270
	-GCN	57.27	65.84	61.26	63.93	44.91	28.45	34.83	30.70	48.05	41.190



(a) (b) (c) (d)

Fig. 6. Results of different DA settings on (a) EF_2, (b) LF_2, (c) MF_1, and (d) mAP. The blue bars represent the DA from UCM to AID, and red bars are scores of the DA from AID to UCM).

dataset to the high-resolution AID dataset, we then conduct the domain adaption from the AID dataset (as the source domain) to the UCM dataset (as the target domain). Same as in Table II, we compare the models designed for two types of tasks.

First, we report the scores of four existing methods for the multilabel image classification task (the MAIC type in Table III), including DELTA [5], KSSNET [7], TResNet [6], and ASL [43]. As noted from Table III, the proposed DA-MAIC model outperforms all competitors on the target AID multilabel dataset on almost all metrics that we adopt in this article. Specifically, DA-MAIC surpasses TResNet 25.63% on EP and 19.16% on LP, respectively. On the F-scores, which are the combination of precision and recall, DA-MAIC improves significantly in comparison with the second place on EF_1 (from 50.83% of DELTA to 77.23%) and LF_1 (from 31.66% of TResNet to 46.13%).

Second, we report the scores of two existing methods for the DA task (the DA type in Table II), including CDAN [44] and MCD [13]. In comparison to MCD, DA-MAIC increases 25.70% on MF_1 score and improves mAP by 4.935%. The quantitative results demonstrate the feasibility of applying the DA mechanism to the MAIC task.

With respect to DA-MAIC, each component is evaluated by the ablation study. The score decreases of -DC and -GCN illustrate the necessity of the domain classifier and the label encoder in this new MAIC task. In summary, the DC mitigates domain shifts across distributions, and the label encoder bridges the divide between the DA methods and MAIC tasks.

In addition, the bar charts show the comparison results of DA between two directions (UCM→AID and AID→UCM). Results of the two directions on DELTA, KSSNet, TResNet, and the proposed DA-MAIC are shown in Fig. 6. The blue bars stand for the direction of UCM→AID, and the red bars are scores of AID→UCM. In general, we have better performance of the AID→UCM than that of the UCM→AID on EF_2, LF_2, and mAP. Especially on EF_1 and LF_1, the AID→UCM is significantly higher than the UCM→AID for each model. For DELTA, the difference between the two directions is less obvious than that of other models. It might be due to the following reason: DELTA aims at extracting more fine-gained features of images, so more task-specific features can be learned in the UCM→AID process, which narrows the gap between two domains. In addition, DA-MAIC still yields the best performance on almost all metrics studied in this article, indicating the effective predicting capability of the proposed model.

As a consequence, the spatial resolution of the source domain is an important property that has a strong impact on the results. It can be easier to transfer from high-quality images to low-quality images. The reason can be the following: 1) more image features of the AID dataset are extracted to mitigate domain shifts across distributions during multiple label predictions and 2) uphill is harder than downhill. The domain gradient (which is the difference of two feature spaces) in AID→UCM is smaller than that in UCM→AID.

3) DA Between Datasets With Similar and Disparate Spatial Resolutions: Base on the results above, we can preliminarily

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TABLE IV
DA RESULTS FROM AID TO DFC15 (%). WE COMPARE WITH SEVERAL EXISTING METHODS FOR TWO TYPES OF TASKS, INCLUDING THE MAIC TASK AND THE DA TASK

Type	Methods	EP	ER	EF_1	EF_2	LP	LR	LF_1	LF_2	MF_1	mAP
MAIC	DELTA [5]	33.80	37.96	35.76	37.05	37.75	51.57	43.60	48.06	39.68	51.590
	KSSNet [7]	28.83	24.42	26.44	25.19	20.05	38.34	26.33	32.42	26.39	45.172
	TResNet [6]	32.42	36.55	34.36	35.64	44.64	55.54	49.49	52.95	41.93	62.360
	ASL [43]	34.90	37.62	36.21	37.04	35.99	40.54	38.13	39.54	37.17	58.262
DA	CDAN [44]	31.67	33.91	32.75	33.44	39.65	51.87	44.94	48.86	38.85	57.526
	MCD [13]	28.56	32.23	30.28	31.42	22.04	50.48	30.69	40.13	30.49	44.413
Ours	**DA-MAIC**	**46.12**	**42.73**	**44.36**	**43.37**	**46.18**	**59.62**	**52.05**	**56.34**	**48.21**	**72.988**

TABLE V
DA RESULTS FROM UCM TO DFC15 (%). WE COMPARE WITH SEVERAL EXISTING METHODS FOR TWO TYPES OF TASKS, INCLUDING THE MAIC TASK AND THE DA TASK

Type	Methods	EP	ER	EF_1	EF_2	LP	LR	LF_1	LF_2	MF_1	mAP
MAIC	DELTA [5]	38.39	13.36	19.83	15.37	19.66	20.16	19.91	20.06	19.87	44.135
	KSSNet [7]	27.57	20.77	23.70	21.85	9.22	33.33	14.44	21.88	19.07	42.996
	TResNet [6]	40.19	16.34	23.23	18.54	35.13	17.80	23.63	19.75	23.43	53.773
	ASL [43]	37.83	24.09	29.43	25.97	39.49	28.51	33.11	30.19	31.27	61.995
DA	CDAN [44]	35.71	24.48	29.05	26.12	25.29	27.84	26.5	27.29	27.78	55.386
	MCD [13]	27.80	23.86	25.68	24.56	19.19	18.97	19.08	19.01	22.38	43.561
Ours	**DA-MAIC**	**43.85**	**25.44**	**32.20**	**27.77**	**42.28**	**33.55**	**37.41**	**34.99**	**34.81**	**68.663**

tell that the spatial resolution is an important property that has a strong impact on the results. To further verify this observation and investigate question 2), we conduct the DA between datasets with similar (AID→DFC15) and disparate (UCM→DFC15) spatial resolutions.

In the DA experiments with DFC15, we consider only the classes in common among DFC15, UCM, and AID datasets, which are water, grass, building, tree, ship, and car. We conduct the DA from UCM to DFC15 and AID to DFC15, respectively, and the results are listed in Tables IV and V. We can see from the two tables that the proposed DA-MAIC model surpasses all competitors on the target DFC15 dataset in not only sample-based indexes but also label-based scores.

a) AID→DFC15: As shown in Table IV, we compare with four existing methods for the multilabel image classification task, including DELTA [5], KSSNET [7], TResNet [6], and ASL [43]. The proposed DA-MAIC model reaches the highest scores on the target DFC15 multilabel dataset. Specifically, on EF_1 and LF_1, DA-MAIC increases by 8.15% and 13.92%, respectively, in comparison with ASL (state-of-the-art). Compared to KSSNet, which resorts to employing the GCN framework for exploring label correlations, our model achieves an improvement of 27.816% on mAP. Comparisons with current MAIC models demonstrate that our model can achieve promising performance for the new dataset. Also, we compare with two existing methods for the DA task, including CDAN [44] and MCD [13]. In comparison to MCD, DA-MAIC increases 17.72% on MF_1 score and improves mAP by 28.575%. The results on AID→DFC15 further demonstrate the advantage of applying the DA mechanism to the MAIC task.

b) UCM→DFC15: As listed in Table V, the DA-MAIC model still outperforms the scores on the target domain DFC15 dataset. DA-MAIC improves by 2.77% and 4.3% on EF_1 and LF_1, in comparison with ASL. On MF_1 and mAP, DA-MAIC reaches 34.81% and 68.663%, surpassing second place by 3.54% and 6.668%. In comparison to MCD, DA-MAIC increases 12.43% on MF_1 and 25.102% on mAP. The results on UCM→DFC15 are constant to the advantage of the proposed DA-MAIC.

Another observation of interest is that the performance of AID→DFC15 (48.21% on MF_1 and 72.988% on mAP) is much higher than that of UCM→DFC15 (34.81% on MF_1 and 68.663% on mAP). This can be due to the similar spatial resolution between AID and DFC15 so that the gap between the two domains is narrowed. To this end, the features learned from the source AID domain can be generalized well on the target DFC15 domain. This observation draws the positive answers: spatial resolution is an important property that has a strong impact on the results.

4) Upper Bound and Lower Bound: To the best of our knowledge, there is no DA model for the MAIC task before. In this way, it is unfair and probably misleading to compare our model with current MAIC models that are trained on large volumes of the labeled dataset and tested directly within the same dataset. Based on this consideration, we carry out the experiments in three settings, the upper bound setting, the lower bound setting, and the Model→DA setting.

1) In the (performance) lower bound setting, it is trained on the source dataset and tested on the target dataset directly without the DA component.

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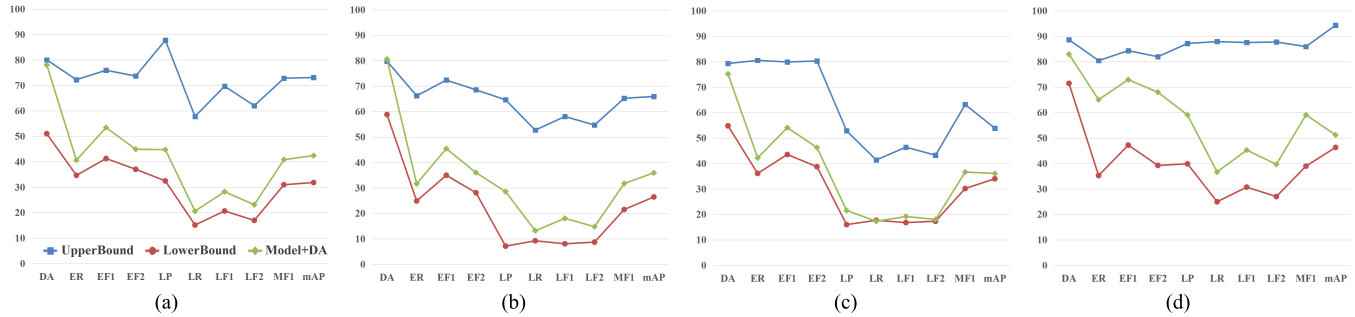
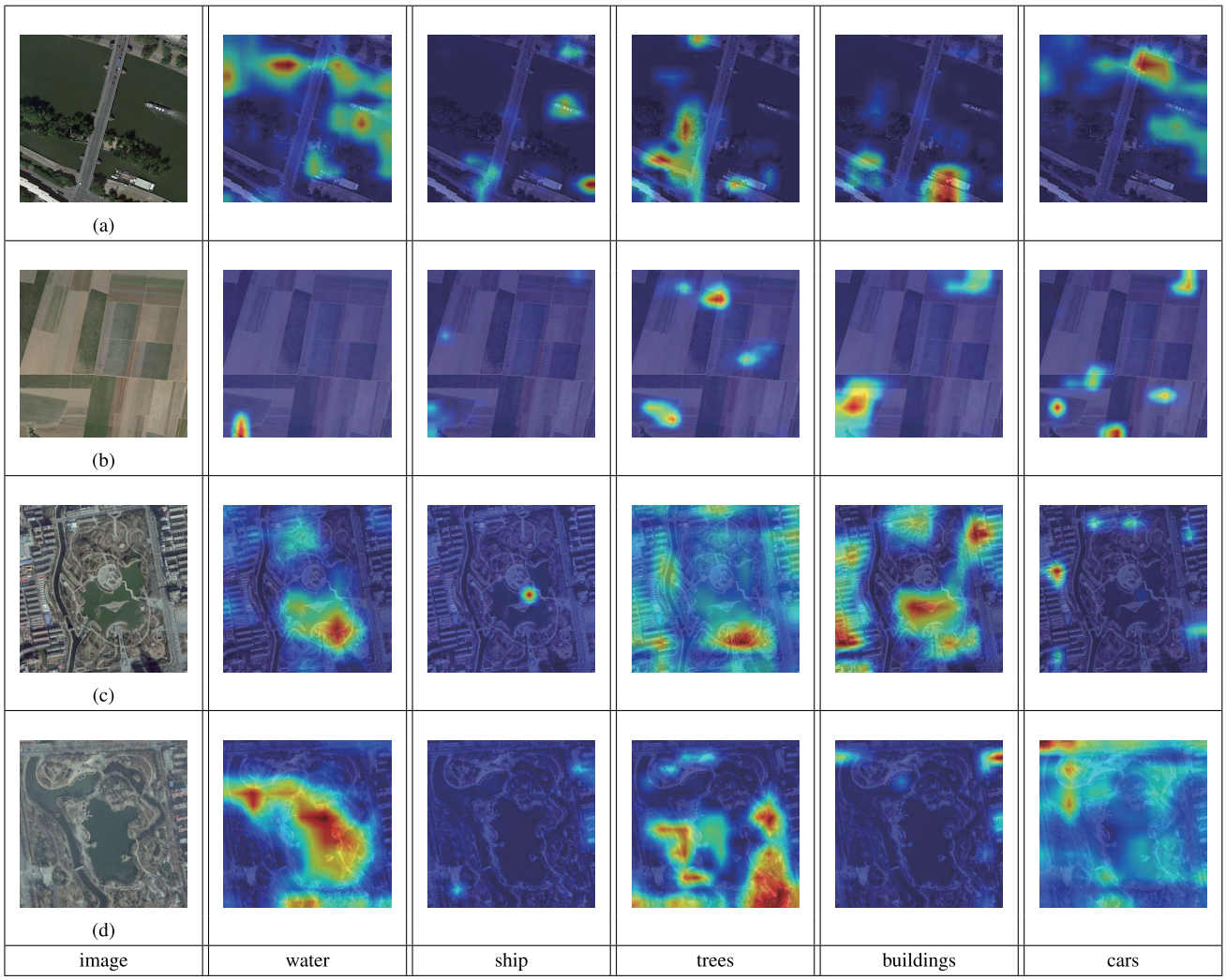


Fig. 7. Under different settings: performance comparisons with the state-of-the-art MAIC models with and without the DA component. (a) DELTA, (b) KSSNet, (c) TRestNet, (d) DA-MAIC (Ours).

TABLE VI

VISUALIZATION FOR LABEL-SPECIFIC FEATURES OF FEW IMAGES SELECTED FROM THE TARGET AID MULTILABEL DATASET, REPRESENTING WATER, SHIP, TREES, BUILDINGS, AND CARS. REGIONS MARKED AS RED ARE STRONGLY ACTIVATED, AND BLUE IMPLIES WEAKLY ACTIVATED



2) In the upper bound setting, it is trained and tested on the same target dataset (divided into annotated training and nonannotated testing sets) without the DA component. This is the same way as in the traditional supervised multilabel image classification task.
3) In the Model→DA setting, we train the model on the annotated source domain and test on the nonannotated target domain with the DA component.

Three settings of experimental results are illustrated in Fig. 7. We set the UCM dataset as the source domain and target on the

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AID dataset. Three current MAIC models (DELTA, KSSNet, and TResnet) together with our proposed model DA-MIAC are compared in four line charts. Intuitively, it can be seen that both sample- and label-based indexes are calculated. In the view of every single model, we can find that the results of the DA setting outperform that of the lower bound setting, but are suboptimal comparing to the upper bound setting. Especially, with the help of the domain classifier, DA-MAIC improves significantly on every index used in this article, as shown in Fig. 7(d). It is clear that, when introducing the DA component into existing methods, the performance improves outstandingly. For example, the performance of UCM-trained DELTA is suboptimal on the AID dataset, but, with the DA component, the numbers grow to a great extent (with the increase of around 10% on MF_1 and mAP). This observation verifies the feasibility of the DA component in the MAIC task.

In addition, for the Model→DA setting, we can tell that our model outperforms the latest models significantly. Specifically, DA-MAIC achieves 82.94% on EP, which is 7.75%, 4.89%, and 2.33% higher than that of TResNet, DELTA, and KSSNet, respectively. Besides, as the combination of EF_1 and LF_1, MF_1 of DA-MAIC is 14.24% higher than the second place (44.85% in TResNet). Also, DA-MAIC gets the highest mAP comparing with the other state of the arts. To summarize, our model achieves the best scores in comparison with all other competitors, which proves the effectiveness and robustness of our method.

C. Qualitative Analysis

To help explain the working of the proposed approach, we have conducted the visualization analysis using gradcam++. We show visualize label-specific features and attentional parcels learned by the DA-MAIC model for the images on the target AID domain. In Table VI, a list of feature regions regarding water, ship, trees, buildings, and cars are organized and illustrated for four example images. To clarify, we select to utilize the most relevantly activated regions as the representative illustration results. Table VI shows that the regions with the most specific semantic information related to the positive labels are activated as red in these feature maps. In contrast, less label-correlated informative patches are weakly activated for those negative labels. For example, the label ship is negative for image (b), and the corresponding visualizations are all blue, which means that no corresponding region is activated. To summarize, the qualitative experiments provide insights and help explain the effectiveness of our model.

D. Comparison of Model Efficiency

We analyze the model efficiency to further evaluate the proposed DA-MAIC model. We concentrate on the calculation complexity and parameters. The floating-point operations (FLOPs) of the methods are calculated using flops_counter package in PyTorch [45], as reported in Table VII. We compare with Resnet-101 and the backbone with only one fully connected layer (Backbone + FC in Table VII), respectively. For the scores on FLOPs, DA-MAIC is 0.09 higher than that of Backbone + FC and 0.08 higher than that of ResNet-101 [46]. For the scores on Parameters, DA-MAIC has 45.47M of parameters, slightly higher than that of Backbone + FC and ResNet-101 [46]. The results of Table VII) show that DA-MAIC does not require much more resources, while it can maintain its effectiveness.

TABLE VII
MODEL EFFICIENCY COMPARISONS

Model	FLOPs ($\times 10^9$)	Parameters ($\times 10^6$)
ResNet-101 [46]	7.85	44.55
Backbone+FC	7.84	43.12
DA-MAIC	7.93	45.47

V. CONCLUSION

In this article, we study the MAIC task and investigate whether the classification model can be learned with no prior label information available. Inspired by the recent DA methods, which can transfer knowledge from the annotated dataset to datasets with no or scarce annotation, we introduce a new research direction for the MAIC task. The proposed DA-based MAIC (DA-MAIC) allows end-to-end training with annotated samples from the source domain and unannotated samples from the target domain. The method mainly consists of two parts: an image classifier is utilized to learn task-specific features based on the GCN to extract label correlation features, and a domain classifier is introduced to align the distributions of features across the two domains. Sufficient experimental results demonstrate that the proposed DA-MAIC can significantly improve the classification performance on the target domain, with domain-invariant features extracted by the DA component.

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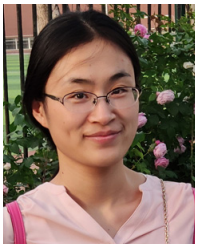
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